

Multiple Choice (10 marks)

1. An automobile travelling at 10 miles per hour produces 0.33 l b of CO gas per mile. How many moles of CO are produced per mile?
 - (a) 5.4 moles per mile
 - (b) 9.0 moles per mile
 - (c) 2.5 moles per mile
 - (d) 7.3 moles per mile
 - (e) 3.8 moles per mile
2. What weight of sulfur must combine with aluminum to form 600 lbs of aluminum sulfide?
 - (a) 350 lbs. sulfur
 - (b) 320 lbs. sulfur
 - (c) 425 lbs. sulfur
 - (d) 400 lbs. sulfur
 - (e) 300 lbs. sulfur
3. Sirius, one of the hottest known stars, has approximately a blackbody spectrum with $\lambda_{\max} = 260$ nm. Estimate the surface temperature of Sirius.
 - (a) 8500 K
 - (b) 15000 K
 - (c) 9500 K
 - (d) 11000 K
4. $\text{SO}_2 + \text{Cl}_2 \rightarrow \text{SO}_2 + \text{Cl}_2$ is 8.0 minutes. In what period of time would the concentration of $\text{SO}_2 + \text{Cl}_2$ be reduced to 1.0% of the original?
 - (a) 43 minutes
 - (b) 103 minutes
 - (c) 93 minutes
 - (d) 113 minutes
 - (e) 23 minutes
5. A gas of molecular weight 58 was found to contain 82.8% C and 17.2% H. What is the molecular formula of this compound?

- (a) C_3 H_6
- (b) C_5 H_10
- (c) C_4 H_8
- (d) C_4 H_9
- (e) C_3 H_8

True / False (10 marks)

Answer True or False

6. True or False: The final concentration of the mixed solution is 0.35 M H_2 SO_4 after combining 50 ml of 0.5 M and 75 ml of 0.25 M H_2 SO_4.
7. True or False: A full CI calculation of CH_2 SiHF using a 6-31 G** basis set results in 4.67×10^{28} configuration state functions.
8. True or False: It is necessary to use 72 g of NH_3 to fully react with 80 g of O_2 based on the balanced chemical equation.
9. True or False: The uncertainty of the momentum of an electron with a position known within a 10 pm interval is approximately $4.2 \times 10^{-23} \text{ kg m s}^{-1}$.
9. True or False: During the boiling of liquid NH_3, the hydrogen bonds between separate NH_3 molecules break apart.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between kinetic energy and wavelength for a photon and an electron, each possessing 1 eV of kinetic energy. Calculate and compare their wavelengths, explaining the implications of your findings.
12. Discuss the weight of 1,000 cubic feet of air at standard temperature and pressure (STP), including the calculations and underlying principles involved in your analysis.

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